

CASE STUDY 2 – SIOP REPORTING

Financial Forecast Performance Dashboard

Power BI Case Study Report

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Tools Used: Power BI | Power Query | DAX

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1. Objective

The objective of this case study is to design and develop an interactive Power BI dashboard that compares forecast values with actual values to evaluate business performance. The dashboard enables business users to monitor Revenue, Non-Revenue, CAPEX, Forecast Accuracy, and Forecast Bias using dynamic filters and executive-level visualizations.

2. Business Requirements

The following business requirements were implemented:

- Forecast Accuracy % = $1 - \text{ABS}(\text{Lags Qty} - \text{Actuals Qty}) / \text{Actuals Qty}$
- Forecast Bias % = $(\text{Lags Qty} - \text{Actuals Qty}) / \text{Actuals Qty}$
- Executive dashboard for Revenue, Non-Revenue and CAPEX Forecast by Country
- Lag Selection filter (Resultant, Lag0, Lag1)
- Country and Period filters
- Dynamic KPI calculations based on selected Lag

3. Data Sources

The dashboard was created using the following datasets:

- Forecast Table
- Actual Table

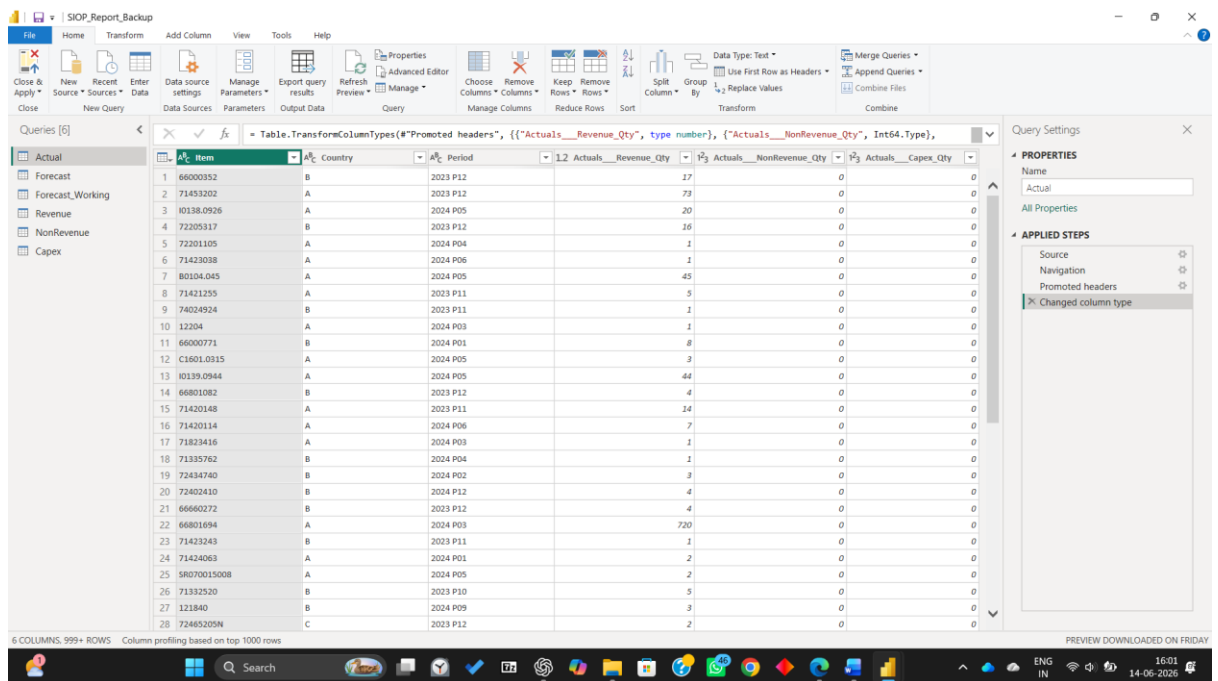
The Forecast table contains forecast values for Revenue, Non-Revenue and CAPEX, while the Actual table contains the corresponding actual values used for KPI calculations.

4. Power Query Transformations

The following transformations were performed in Power Query:

1. Imported Forecast and Actual datasets
2. Created Lag column using AsOfPeriod
3. Removed unnecessary columns
4. Grouped forecast data
5. Pivoted Lag values
6. Merged Forecast and Actual tables
7. Loaded transformed data into Power BI

Power Query Editor



The screenshot displays the Power Query Editor window for a query named "Table.TransformColumnTypes(*Promoted headers*, {{'Actuals__Revenue_Qty', type number}, {'Actuals__NonRevenue_Qty', Int64.Type}, {'Actuals__Capex_Qty', Int64.Type}})". The data table has 6 columns and 999 rows. The columns are: Item, Country, Period, Actuals__Revenue_Qty, Actuals__NonRevenue_Qty, and Actuals__Capex_Qty. The data is organized into a table with 28 rows and 6 columns. The first column is 'Item', followed by 'Country', 'Period', and three columns for financial metrics: 'Actuals__Revenue_Qty', 'Actuals__NonRevenue_Qty', and 'Actuals__Capex_Qty'. The 'Actuals__Revenue_Qty' column contains values ranging from 1 to 720. The 'Actuals__NonRevenue_Qty' and 'Actuals__Capex_Qty' columns contain values ranging from 0 to 7. The 'Period' column contains values like '2023 P12', '2024 P01', '2024 P05', '2024 P11', '2024 P12', '2024 P03', '2024 P06', '2024 P09', and '2024 P10'. The 'Country' column contains values 'A', 'B', and 'C'. The 'Item' column contains various alphanumeric codes. The right-hand pane shows the 'Query Settings' for the 'Actual' query, with the 'Name' field set to 'Actual'. The 'APPLIED STEPS' list includes 'Source', 'Navigation', 'Promoted headers', and 'Changed column type'. The bottom status bar indicates '6 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Item	Country	Period	Actuals__Revenue_Qty	Actuals__NonRevenue_Qty	Actuals__Capex_Qty
66000352	B	2023 P12	27	0	0
71453202	A	2023 P12	73	0	0
10138.0926	A	2024 P05	20	0	0
72205317	B	2023 P12	16	0	0
72301105	A	2024 P04	1	0	0
71423038	A	2024 P06	1	0	0
80104.045	A	2024 P05	45	0	0
71421255	A	2023 P11	5	0	0
74024924	B	2023 P11	1	0	0
12204	A	2024 P03	1	0	0
66000771	B	2024 P01	8	0	0
C1601.0315	A	2024 P05	3	0	0
10139.0944	A	2024 P05	44	0	0
66801082	B	2023 P12	4	0	0
71420148	A	2023 P11	14	0	0
71420114	A	2024 P06	7	0	0
71823416	A	2024 P03	1	0	0
71335762	B	2024 P04	1	0	0
72434740	B	2024 P02	3	0	0
72402410	B	2024 P12	4	0	0
66660272	B	2023 P12	4	0	0
66801694	A	2024 P03	720	0	0
71423243	B	2023 P11	1	0	0
71424063	A	2024 P01	2	0	0
58070015008	A	2024 P05	2	0	0
71332520	B	2023 P10	5	0	0
121840	B	2024 P09	3	0	0
72465205N	C	2023 P12	2	0	0

Applied Steps

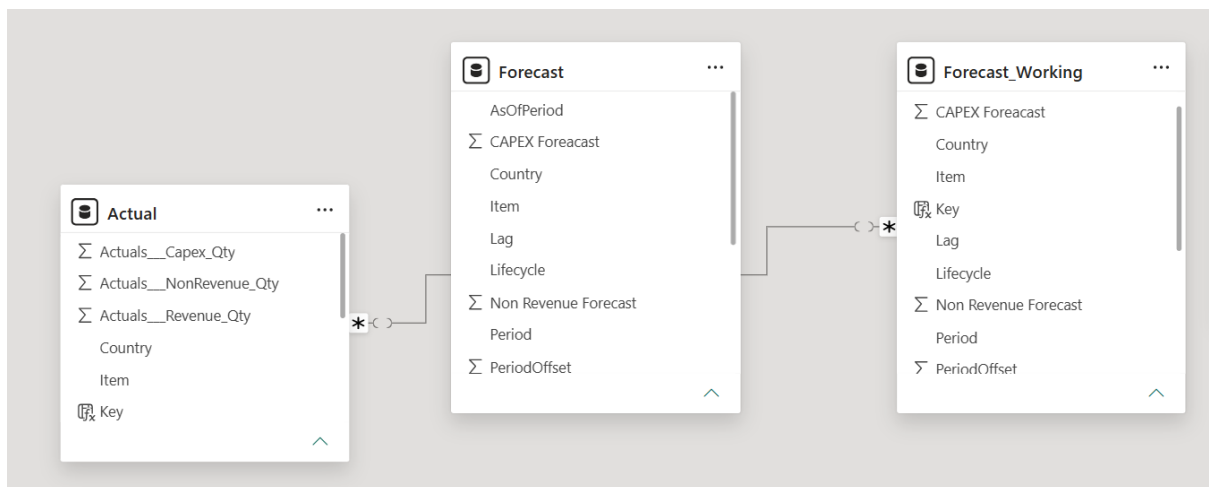
Table: ExpandTableColumn(#Merged Queries2", "Actual", {"Actuals__Revenue_Qty", "Actuals__NonRevenue_Qty", "Actuals__Capex_Qty"},

Item	Country	Period	Lag1 Revenue	Resultant Revenue	Lag0 Revenue	Lag1 Non Revenue
71453202	A	2023 P12	77.59	24.45	3	
72205317	B	2023 P12	10	null	2	
72205317	B	2023 P12	10	null	2	
72205317	B	2023 P12	10	null	2	
72205317	B	2023 P12	10	null	2	
72205317	B	2023 P12	10	null	2	
10-614-25	A	2024 P05	2.76	null	null	
10-614-25	A	2024 P05	2.76	null	null	
10-614-25	A	2024 P05	2.76	null	null	
10-614-25	A	2024 P05	2.76	null	null	
71423038	A	2024 P06	2	null	null	
71421255	A	2023 P11	3.08	null	4	
71421255	A	2023 P11	3.08	null	4	
71421255	A	2023 P11	3.08	null	4	
71421255	A	2023 P11	3.08	null	4	
71421255	A	2023 P11	3.08	null	4	
66801082	B	2023 P12	null	2.56	null	
66801082	B	2023 P12	null	2.56	null	
66801082	B	2023 P12	null	2.56	null	
71420148	A	2023 P11	49	null	null	
71420148	A	2023 P11	49	null	null	
71420148	A	2023 P11	49	null	null	
71420148	A	2023 P11	49	null	null	
71420148	A	2023 P11	49	null	null	
10-800-12	A	2024 P06	null	3	11	
71420114	A	2024 P06	0.88	null	1.3	
71420114	A	2024 P06	0.88	null	1.3	

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON FRIDAY

Data Model



5. DAX Measures

The following DAX measures were created to support dynamic KPI calculations:

- Selected Revenue

```
1 Selected Revenue =  
2 VAR SelectedLag =  
3     SELECTEDVALUE('Lag Selection'[Lag], "Lag0")  
4 RETURN  
5     SWITCH(  
6         TRUE(),  
7         SelectedLag = "Resultant", SUM(Revenue[Resultant Revenue]),  
8         SelectedLag = "Lag0", SUM(Revenue[Lag0 Revenue]),  
9         SelectedLag = "Lag1", SUM(Revenue[Lag1 Revenue])  
10    )
```

- Selected Non-Revenue

Line	Formula	Properties	Calc
1	Selected NonRevenue =		
2	VAR SelectedLag =		
3	SELECTEDVALUE('Lag Selection'[Lag], "Lag0")		
4	RETURN		
5	SWITCH(6 TRUE(), 7 SelectedLag = "Resultant", SUM(Revenue[Resultant Non Revenue]), 8 SelectedLag = "Lag0", SUM(Revenue[Lag0 Non Revenue]), 9 SelectedLag = "Lag1", SUM(Revenue[Lag1 Non Revenue]), 10 BLANK() 11)		

- Selected CAPEX

```
1 Selected Capex =  
2 VAR SelectedLag =  
3     SELECTEDVALUE('Lag Selection'[Lag], "Lag0")  
4 RETURN  
5     SWITCH(  
6         TRUE(),  
7         SelectedLag = "Resultant", SUM(Revenue[Resultant Capex]),  
8         SelectedLag = "Lag0", SUM(Revenue[Lag0 Capex]),  
9         SelectedLag = "Lag1", SUM(Revenue[Lag1 Capex]),  
10        BLANK()  
11    )
```

- Actual Revenue

```
1 Actual Revenue =  
2 SUM(Revenue[Actuals__Revenue_Qty])
```

- Forecast Accuracy %

```
1 Forecast Accuracy % =  
2 1 -  
3 DIVIDE(  
4     ABS([Selected Revenue] - [Actual Revenue]),  
5     [Actual Revenue]  
6 )
```

- Forecast Bias %

```
1 Forecast Bias % =  
2 DIVIDE(  
3     [Selected Revenue] - [Actual Revenue],  
4     [Actual Revenue]  
5 )
```

6. Dashboard Overview

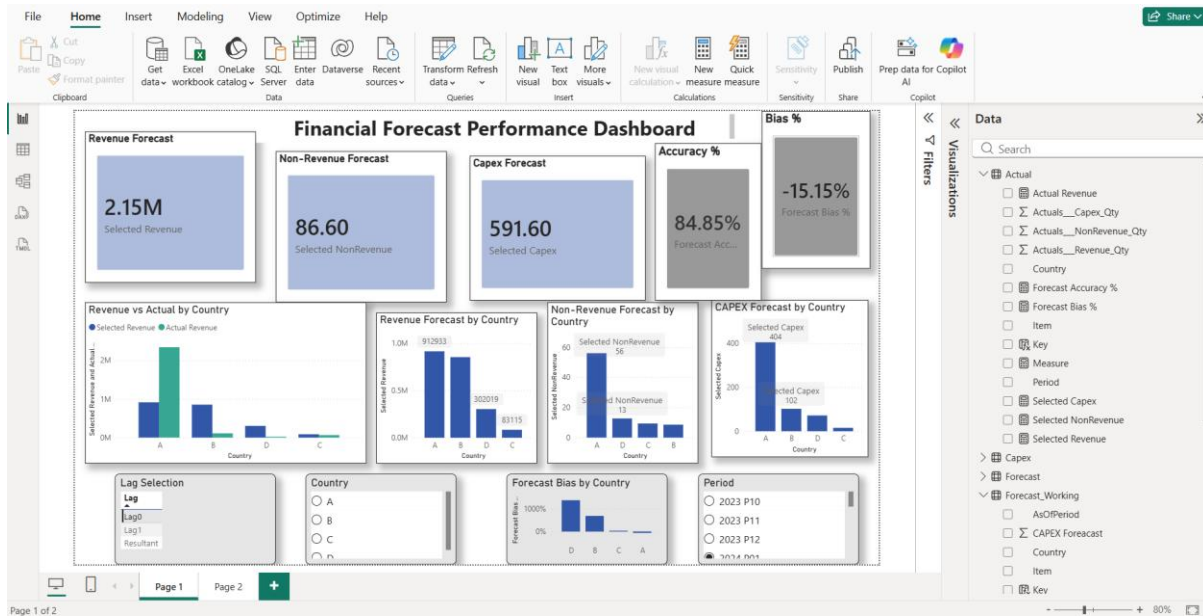
The executive dashboard provides an interactive view of forecast performance across Revenue, Non-Revenue and CAPEX.

The dashboard contains:

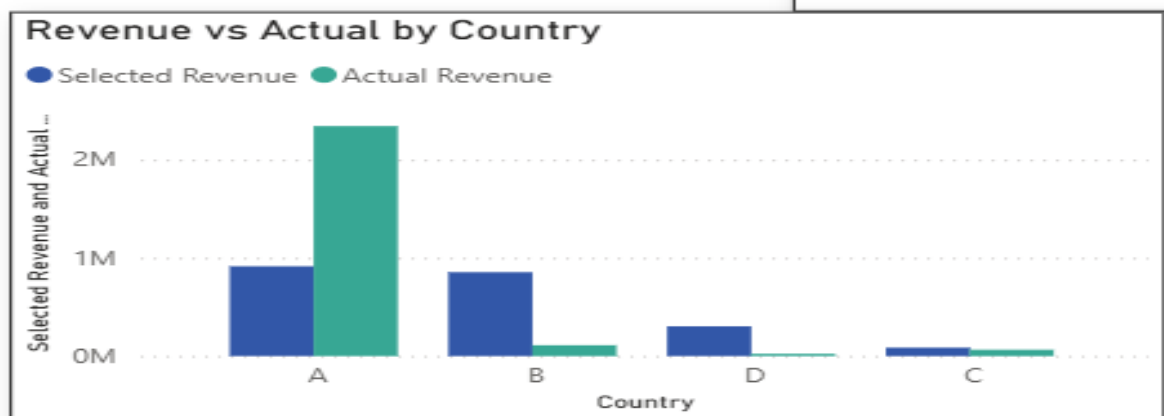
- Revenue KPI Card
- Non-Revenue KPI Card
- CAPEX KPI Card
- Forecast Accuracy KPI
- Forecast Bias KPI
- Revenue Forecast vs Actual Revenue
- Revenue Forecast by Country
- Non-Revenue Forecast by Country
- CAPEX Forecast by Country
- Forecast Bias by Country
- Lag Selection Slicer

- Country Slicer
- Period Slicer

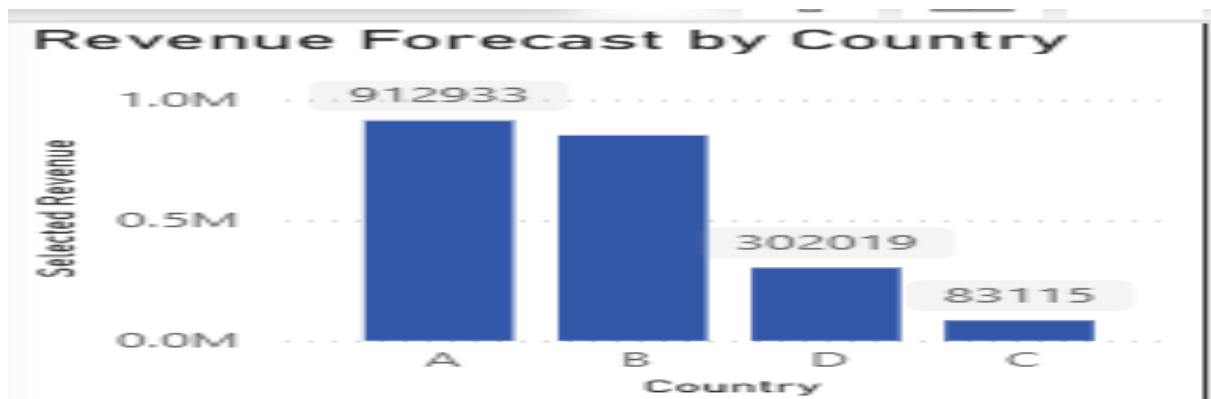
Complete Dashboard



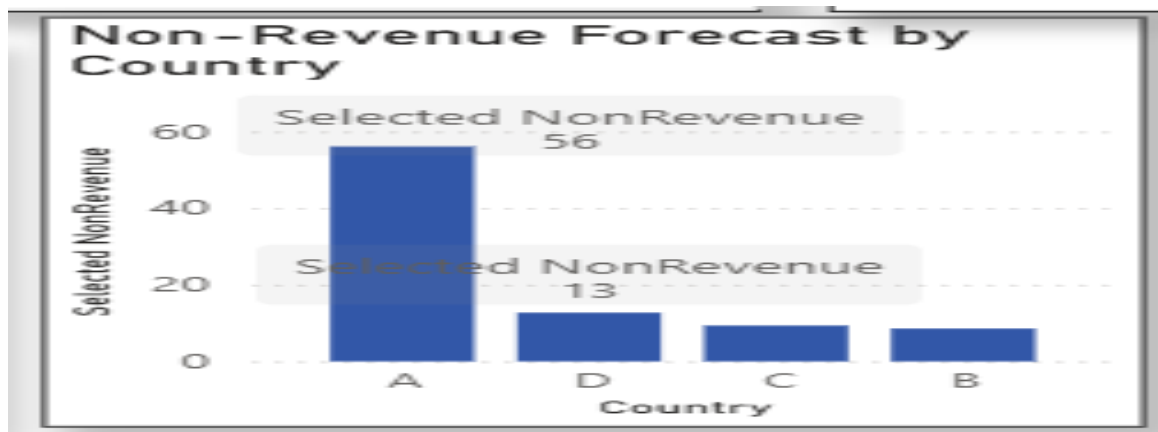
Revenue Forecast vs Actual Revenue



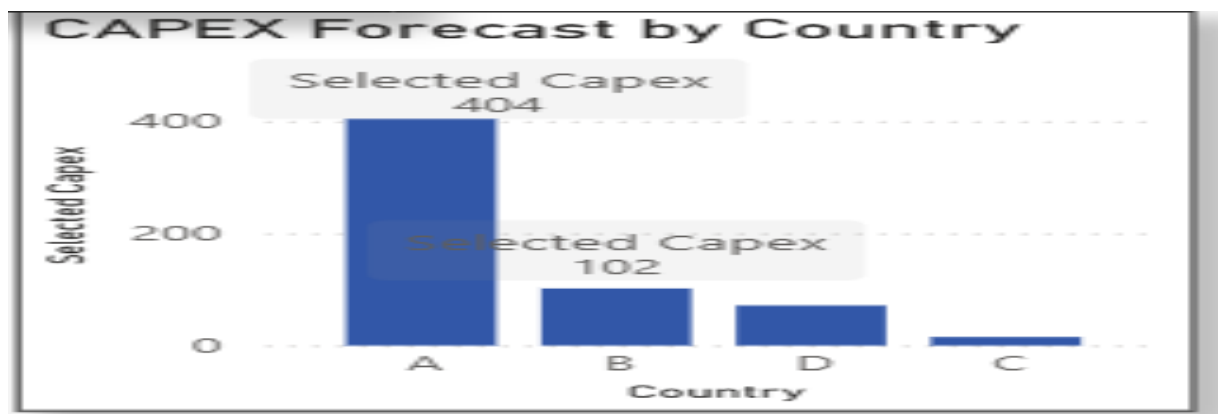
Revenue Forecast by Country



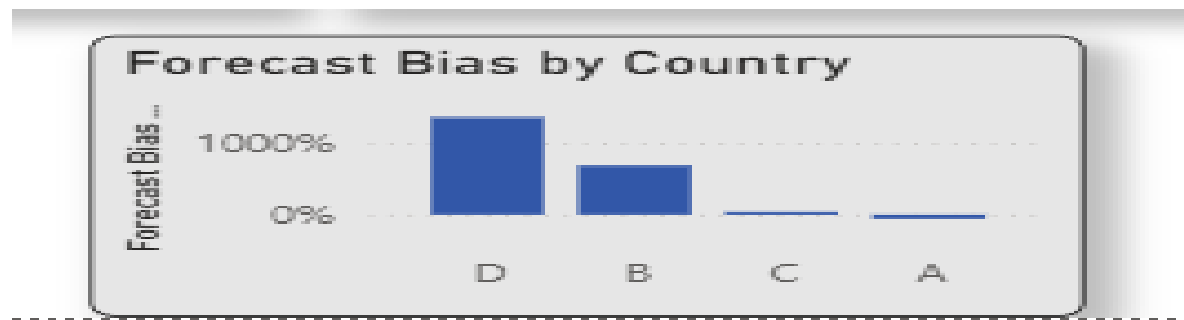
Non-Revenue Forecast by Country



CAPEX Forecast by Country



Forecast Bias by Country



Lag Selection Filter and Country Filter

Lag Selection

☒ Lag

☐ Lag0

☐ Lag1

☐ Resultant

Country

☐ A

☐ B

☐ C

☐ D

Period Filter

Period

☐ 2023 P10

☐ 2023 P11

☐ 2023 P12

☒ 2024 P01

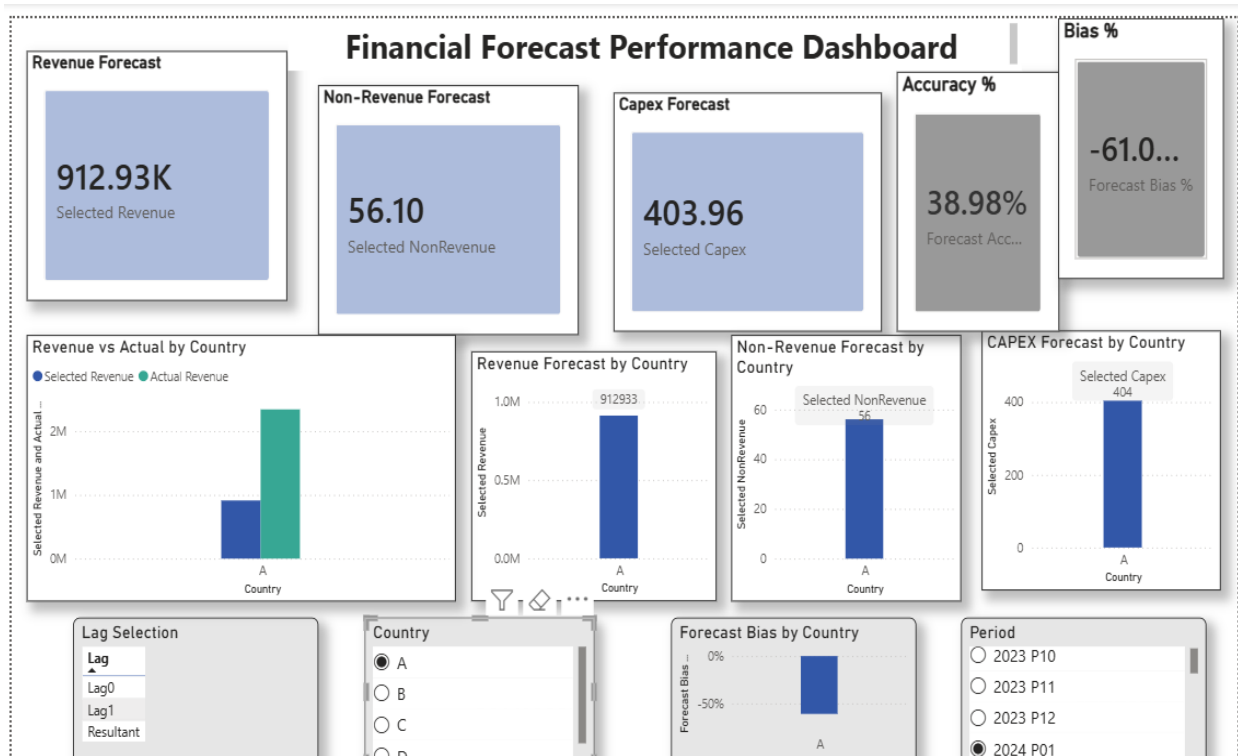
7. After Dashboard Overview

Interactive Dashboard Features

The dashboard is fully interactive. Users can analyze forecast performance by applying filters for Country, Period, and Lag Selection. All KPI cards and charts automatically update based on the selected filter values, enabling flexible and detailed business analysis

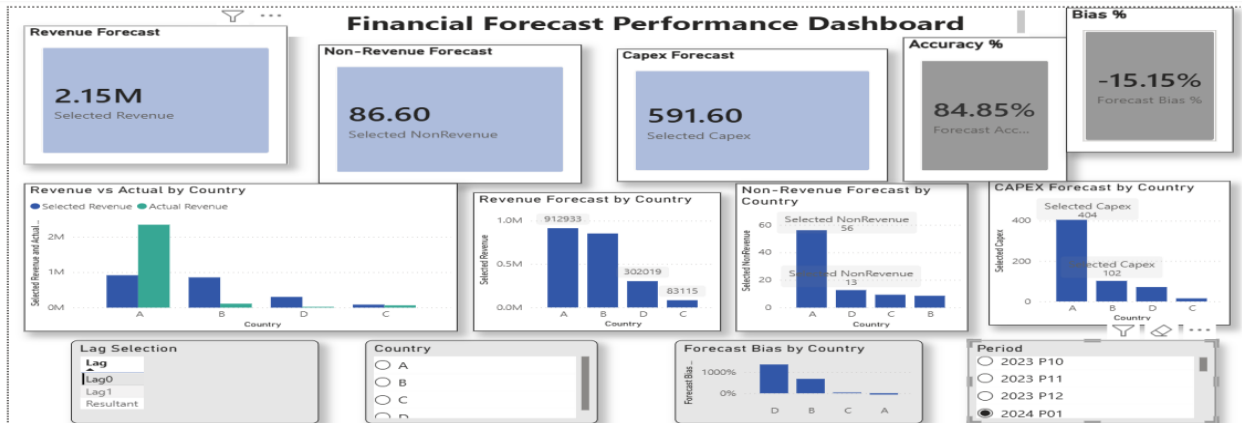
i. Country Filter Applied (Country = A)

After selecting Country A, all KPI cards and visualizations automatically update to display data for the selected country



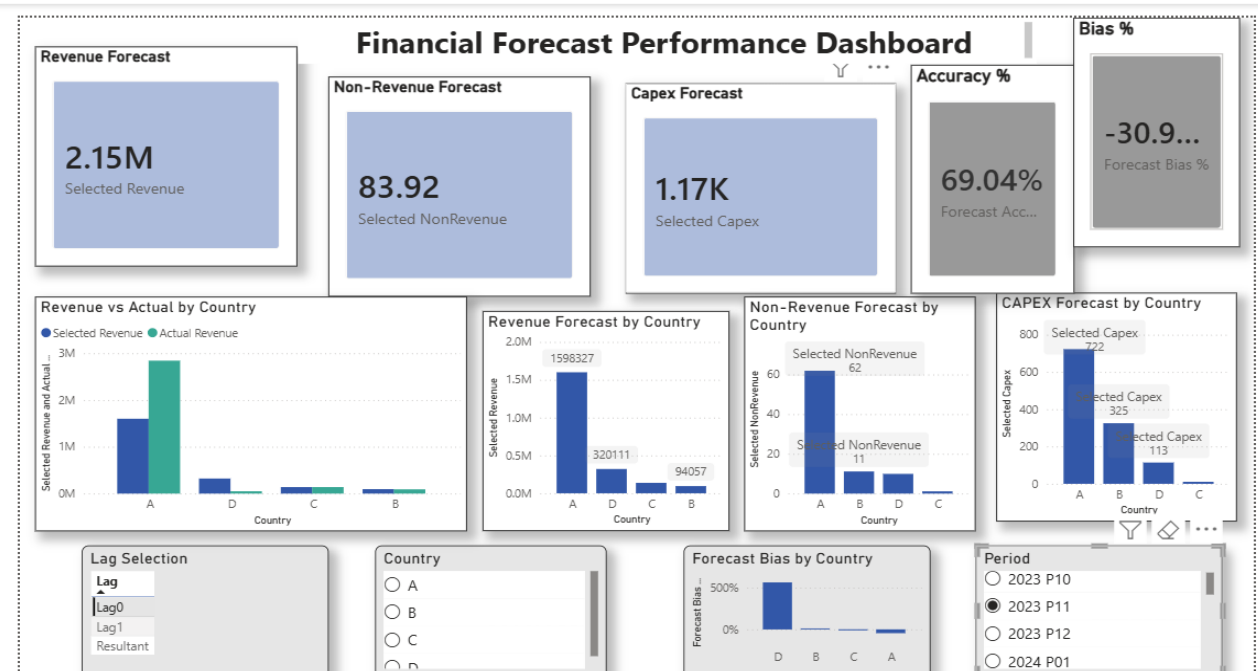
ii. Lag Selection Changed (Lag0)

Selecting **Lag0** updates Revenue, Non-Revenue, CAPEX, Forecast Accuracy, and Forecast Bias based on the Lag0 forecast values.



iii. Period Filter Applied (2023 P11)

Applying the Period filter allows users to analyze forecast performance for a specific reporting period.



8. Key Insights

The developed dashboard provides the following business insights:

- Revenue forecast performance varies across different countries.
- Forecast Accuracy changes dynamically based on the selected Lag.
- Forecast Bias helps identify over-forecasting and under-forecasting.
- Country and Period slicers enable focused analysis.
- Executive users can monitor Revenue, Non-Revenue and CAPEX performance interactively.
- Dynamic KPIs provide quick visibility into forecast quality and business performance.

9. Conclusion

The Power BI dashboard successfully transforms forecast and actual data into an interactive executive reporting solution. Using Power Query transformations, DAX measures and interactive visualizations, the dashboard enables users to monitor Revenue, Non-Revenue, CAPEX, Forecast Accuracy and Forecast Bias through dynamic filtering. The solution provides valuable business insights and supports data-driven decision-making.